

Literature Status Note for arXiv:math/0512207

Run fa_banach_001

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Status. `literature_already_answered`. This note records a later-literature answer to the Slicing Problem signal in Emanuel Milman’s *Dual Mixed Volumes and the Slicing Problem*, arXiv:math/0512207. It is not a new proof from this run.

Source Question

The source paper is E. Milman, *Dual Mixed Volumes and the Slicing Problem*, arXiv:math/0512207. In the Introduction, pages 1–2 of the arXiv PDF, Milman recalls the Slicing Problem. In the notation of that paper, the question is whether there is a universal constant C such that

$$L_K \leq C$$

for every centrally symmetric convex body $K \subset \mathbb{R}^n$, after putting K in isotropic position and normalizing its volume to one. Milman also gives Bourgain’s equivalent hyperplane-section formulation: every centrally symmetric volume-one convex body should have an $(n - 1)$ -dimensional section whose volume is bounded below by a universal positive constant.

Supporting Answer

The supporting paper is B. Klartag and J. Lehec, *Affirmative Resolution of Bourgain’s Slicing Problem using Guan’s Bound*, arXiv:2412.15044. On page 1, the abstract states that the paper gives the final affirmative step for Bourgain’s slicing problem. The theorem asserted there is that for every convex body $K \subseteq \mathbb{R}^n$ of volume one, there exists a hyperplane H and a universal constant $c > 0$ such that

$$\text{Vol}_{n-1}(K \cap H) > c.$$

This is precisely the hyperplane-section formulation recalled in the source paper, and it is stronger in scope than the centrally symmetric case mentioned there. Therefore the source’s Slicing Problem signal is now answered by later literature.

Why This Is Not A New Result

The supporting authors explicitly identify their theorem as an affirmative resolution of Bourgain’s slicing problem. The relation is therefore an exact later-literature answer, not an agent-derived proof, counterexample, or hidden implication.

Scope Limitations

This packet records only the literature status of the Slicing Problem/hyperplane conjecture signal in arXiv:math/0512207. It does not address the separate question in the source about whether $BP_k^n = I_k^n$. It also does not address the later radial-sum questions or the outer mean-radius assumptions for the cube and for unconditional bodies. Those remain separate mathematical targets.

Search And Verification Notes

The run indexes were searched for the source arXiv id, title terms, and core keywords including `slicing`, `hyperplane conjecture`, `dual mixed volumes`, `BP_k`, and `radial sums`. No exact packet for arXiv:math/0512207 was found. A related prior packet for a different source, arXiv:1310.1204, already recorded arXiv:2412.15044 as the decisive answer to Bourgain's slicing problem; its supporting PDF is copied into this packet.

Local Files

- Source PDF: `source_paper.pdf`.
- Supporting PDF: `supporting_paper_2412.15044.pdf`.
- Ledger record: `0512207_slicing_answered_by_2412.15044.json`.

References

- [1] E. Milman, *Dual Mixed Volumes and the Slicing Problem*, arXiv:math/0512207.
- [2] B. Klartag and J. Lehec, *Affirmative Resolution of Bourgain's Slicing Problem using Guan's Bound*, arXiv:2412.15044.